

```
mysql> INSERT INTO Employees VALUES (106,'Lohith',3,400000,103, 'Lohith@kluniv.in');
Query OK, 1 row affected (0.03 sec)
```

```
mysql> SELECT * FROM Employees;
```

EmpID	Name	DeptID	Salary	ManagerID	Email
101	Ravi	1	45000.00	NULL	ravi@kluniv.in
102	Rahul	1	55000.00	101	rahul@kluniv.in
103	Sneha	2	30000.00	101	sneha@kluniv.in
104	Anil	3	70000.00	102	anil@kluniv.in
105	Priya	NULL	40000.00	102	priya@kluniv.in
106	Lohith	3	400000.00	103	Lohith@kluniv.in

```
6 rows in set (0.00 sec)
```

```
mysql> update employees set salary = salary + (salary * 0.1);
```

```
Query OK, 6 rows affected (0.03 sec)
Rows matched: 6 Changed: 6 Warnings: 0
```

```
mysql> SELECT * FROM Employees;
```

EmpID	Name	DeptID	Salary	ManagerID	Email
101	Ravi	1	49500.00	NULL	ravi@kluniv.in
102	Rahul	1	60500.00	101	rahul@kluniv.in
103	Sneha	2	33000.00	101	sneha@kluniv.in
104	Anil	3	77000.00	102	anil@kluniv.in
105	Priya	NULL	44000.00	102	priya@kluniv.in
106	Lohith	3	440000.00	103	Lohith@kluniv.in

```
6 rows in set (0.00 sec)
```

```
mysql> delete from employees where salary < 30000;
```

```
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> INSERT INTO Employees (EmpID, Name, DeptID, Salary, ManagerID, Email) VALUES;
```

```
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near ''
at line 1
```

```
mysql> INSERT INTO Employees (EmpID, Name, DeptID, Salary, ManagerID, Email) VALUES(101, 'Ravi', 1, 45000, NULL, 'ravi@kluniv.in'),(102, 'Rahul', 1, 55000,
101, 'rahul@kluniv.in'),(103, 'Sneha', 2, 30000, 101, 'sneha@kluniv.in'),(104, 'Anil', 3, 70000, 102, 'anil@kluniv.in'),(105, 'Priya', NULL, 40000, 102, 'pr
iya@kluniv.in');
```

```
ERROR 1452 (23000): Cannot add or update a child row: a foreign key constraint fails ('practice2'.employees, CONSTRAINT 'employees_ibfk_1' FOREIGN KEY ('D
```

```
eptID') REFERENCES 'departments' ('DeptID'))
```

```
mysql> INSERT INTO Departments (DeptID, DeptName) VALUES
```

```
    -> (1, 'CSE'),
```

```
    -> (2, 'ECE'),
```

```
    -> (3, 'IT'),
```

```
    -> (4, 'MECH');
```

```
Query OK, 4 rows affected (0.03 sec)
```

```
Records: 4 Duplicates: 0 Warnings: 0
```

```
mysql> INSERT INTO Employees (EmpID, Name, DeptID, Salary, ManagerID, Email) VALUES
```

```
    -> (101, 'Ravi', 1, 45000, NULL, 'ravi@kluniv.in');
```

```
    -> (102, 'Rahul', 1, 55000, 101, 'rahul@kluniv.in');
```

```
    -> (103, 'Sneha', 2, 30000, 101, 'sneha@kluniv.in');
```

```
    -> (104, 'Anil', 3, 70000, 102, 'anil@kluniv.in');
```

```
    -> (105, 'Priya', NULL, 40000, 102, 'priya@kluniv.in');
```

```
Query OK, 5 rows affected (0.04 sec)
```

```
Records: 5 Duplicates: 0 Warnings: 0
```

```
mysql> SELECT * FROM Employees;
```

EmpID	Name	DeptID	Salary	ManagerID	Email
101	Ravi	1	45000.00	NULL	ravi@kluniv.in
102	Rahul	1	55000.00	101	rahul@kluniv.in
103	Sneha	2	30000.00	101	sneha@kluniv.in
104	Anil	3	70000.00	102	anil@kluniv.in
105	Priya	NULL	40000.00	102	priya@kluniv.in

```
5 rows in set (0.01 sec)
```

```
mysql> INSERT INTO Employees (EmpID, Name, DeptID, Salary, ManagerID, Email) VALUES
```

```
    -> (106, 'Lohith', 6, 400000, 103, 'Lohith@kluniv.in');
```

```
ERROR 1452 (23000): Cannot add or update a child row: a foreign key constraint fails ('practice2'.employees, CONSTRAINT 'employees_ibfk_1' FOREIGN KEY ('D
```

```
eptID') REFERENCES 'departments' ('DeptID'))
```

```
mysql> INSERT INTO Employees VALUES (106,'Lohith',6,400000,103, 'Lohith@kluniv.in');
```

```
ERROR 1452 (23000): Cannot add or update a child row: a foreign key constraint fails ('practice2'.employees, CONSTRAINT 'employees_ibfk_1' FOREIGN KEY ('D
```

```
mysql> INSERT INTO Employees (EmpID, Name, DeptID, Salary, ManagerID, Email) VALUES;
```

```
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near ''
at line 1
```

```
mysql> INSERT INTO Employees (EmpID, Name, DeptID, Salary, ManagerID, Email) VALUES(101, 'Ravi', 1, 45000, NULL, 'ravi@kluniv.in'),(102, 'Rahul', 1, 55000,
101, 'rahul@kluniv.in'),(103, 'Sneha', 2, 30000, 101, 'sneha@kluniv.in'),(104, 'Anil', 3, 70000, 102, 'anil@kluniv.in'),(105, 'Priya', NULL, 40000, 102, 'pr
iya@kluniv.in');
```

```
ERROR 1452 (23000): Cannot add or update a child row: a foreign key constraint fails ('practice2'.employees, CONSTRAINT 'employees_ibfk_1' FOREIGN KEY ('D
```

```
eptID') REFERENCES 'departments' ('DeptID'))
```

```
mysql>
```

```

at line 1
mysql> CREATE TABLE Departments (DeptID INT PRIMARY KEY, DeptName VARCHAR(50));
Query OK, 0 rows affected (0.17 sec)

mysql> CREATE TABLE Employees (EmpID INT PRIMARY KEY, Name VARCHAR(50), DeptID INT, Salary DECIMAL(10,2), ManagerID INT, Email VARCHAR(100), FOREIGN KEY (DeptID)
REFERENCES Departments(DeptID), FOREIGN KEY (ManagerID) REFERENCES Employees(EmpID));
Query OK, 0 rows affected (0.13 sec)

mysql> CREATE TABLE Projects (ProjectID INT PRIMARY KEY, ProjectName VARCHAR(100), DeptID INT, FOREIGN KEY (DeptID) REFERENCES Departments(DeptID));
Query OK, 0 rows affected (0.09 sec)

mysql> CREATE TABLE Customers (CustID INT PRIMARY KEY, CustName VARCHAR(100), Email VARCHAR(100));
Query OK, 0 rows affected (0.05 sec)

mysql> CREATE TABLE Orders (OrderID INT PRIMARY KEY, CustID INT, OrderDate DATE, TotalAmount DECIMAL(10,2), FOREIGN KEY (CustID) REFERENCES Customers(CustID));
Query OK, 0 rows affected (0.12 sec)

mysql> CREATE TABLE Products (ProductID INT PRIMARY KEY, ProductName VARCHAR(100), Price DECIMAL(10,2));
Query OK, 0 rows affected (0.07 sec)

mysql> CREATE TABLE Order_Details (OrderDetailID INT PRIMARY KEY, OrderID INT, ProductID INT, Quantity INT, FOREIGN KEY (OrderID) REFERENCES Orders(OrderID), FOR
EIGN KEY (ProductID) REFERENCES Products(ProductID));
Query OK, 0 rows affected (0.12 sec)

```

```

mysql> use practice2;
Database changed
mysql> show tables;
Empty set (0.05 sec)

mysql> ;
ERROR:
No query specified

mysql> show databases;
+-----+
| Database |
+-----+
| bookflow_db |
| information_schema |
| mysql |
| performance_schema |
| practice1 |
| practice2 |
| studentdb1 |
| sys |
+-----+
8 rows in set (0.00 sec)

```

```

mysql> select lower(c.custname)as custname,
-> sum(od.quantity*p.price)as totalpurchase,
-> rank()over(order by sum(od.quantity*p.price)desc)as rank
-> from customers c
-> join orders o on c.custid=o.custid
-> join order_details od on o.orderid=od.orderid
-> join products p on od.productid=p.productid
-> group by c.custname;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'r
ank
from customers c
join orders o on c.custid=o.custid
join order_details od' at line 3
mysql> select lower(c.custname)as custname,
-> sum(od.quantity*p.price)as totalpurchase,
-> rank()over(order by sum(od.quantity*p.price)desc)as rank
-> from customers c
-> join orders o on c.custid=o.custid
-> join order_details od on o.orderid=od.orderid
-> join products p on od.productid=p.productid
-> group by c.custname;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'r
ank
from customers c
join orders o on c.custid=o.custid
join order_details od o' at line 3
mysql> select lower(c.custname)as custname,
-> sum(od.quantity*p.price)as totalpurchase
-> from customers c
-> join orders o on c.custid=o.custid
-> join order_details od on o.orderid=od.orderid
-> join products p on od.productid=p.productid
-> group by c.custname
-> order by totalpurchase desc;
Empty set (0.01 sec)

```

```

mysql> select lower(c.custname)as custname
-> from customers c
-> where not exists(select 1 from orders o where o.custid=c.custid);
Empty set (0.05 sec)

mysql> select lower(c.custname)as custname
-> from customers c
-> where not exists(select 1 from orders o where o.custid=c.custid);
Empty set (0.00 sec)

mysql> select lower(p.productname)as productname
-> from products p
-> join order_details od on p.productid=od.productid
-> group by p.productname
-> order by sum(od.quantity)desc
-> limit 1;
Empty set (0.03 sec)

mysql> select lower(c.custname)as custname,
-> sum(od.quantity*p.price)as totalpurchase,
-> rank()over(order by sum(od.quantity*p.price)desc)as rank
-> from customers c
-> join orders o on c.custid=o.custid
-> join order_details od on o.orderid=od.orderid
-> join products p on od.productid=p.productid
-> group by c.custname;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'r
ank
from customers c
join orders o on c.custid=o.custid
join order_details od' at line 3
mysql> select lower(c.custname)as custname,
-> sum(od.quantity*p.price)as totalpurchase,
-> rank()over(order by sum(od.quantity*p.price)desc)as rank
-> from customers c
-> join orders o on c.custid=o.custid
-> join order_details od on o.orderid=od.orderid
-> join products p on od.productid=p.productid
-> group by c.custname;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'r
ank
from customers c
join orders o on c.custid=o.custid

```

```

mysql> SELECT DeptID FROM Employees GROUP BY DeptID HAVING COUNT(*) > 5;
Empty set (0.00 sec)

mysql> SELECT EmpName, DeptID
-> FROM Employees E
-> WHERE Salary > (SELECT AVG(Salary) FROM Employees WHERE DeptID = E.DeptID);
ERROR 1054 (42S22): Unknown column 'EmpName' in 'field list'
mysql> SELECT Name, DeptID FROM Employees E WHERE Salary > (SELECT AVG(Salary) FROM Employees WHERE DeptID = E.DeptID);

```

```

+-----+-----+
| Name | DeptID |
+-----+-----+
| Rahul | 1 |
| Lohith | 3 |
+-----+-----+
2 rows in set (0.00 sec)

```

```

mysql> select lower(customername)as customername from customers c where not exists(select 1 from orders o where o.customerid=c.customerid);
ERROR 1054 (42S22): Unknown column 'customername' in 'field list'
mysql> select * from employees;

```

```

+-----+-----+-----+-----+-----+-----+
| EmpID | Name | DeptID | Salary | ManagerID | Email |
+-----+-----+-----+-----+-----+-----+
| 101 | Ravi | 1 | 49500.00 | NULL | ravi@kluniv.in |
| 102 | Rahul | 1 | 60500.00 | 101 | rahul@kluniv.in |
| 103 | Sneha | 2 | 33000.00 | 101 | sneha@kluniv.in |
| 104 | Anil | 3 | 77000.00 | 102 | anil@kluniv.in |
| 105 | Priya | NULL | 44000.00 | 102 | priya@kluniv.in |
| 106 | Lohith | 3 | 440000.00 | 103 | Lohith@kluniv.in |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)

```

```
mysql> select * from employees;
```

EmpID	Name	DeptID	Salary	ManagerID	Email
101	Ravi	1	49500.00	NULL	ravi@kluniv.in
102	Rahul	1	60500.00	101	rahul@kluniv.in
103	Sneha	2	33000.00	101	sneha@kluniv.in
104	Anil	3	77000.00	102	anil@kluniv.in
105	Priya	NULL	44000.00	102	priya@kluniv.in
106	Lohith	3	440000.00	103	Lohith@kluniv.in

```
6 rows in set (0.00 sec)
```

```
mysql> select Name,salary from employees where salary >(select avg(salary)from employees);
```

Name	salary
Lohith	440000.00

```
1 row in set (0.00 sec)
```

```
mysql> SELECT EmpName FROM Employees WHERE DeptID = (SELECT DeptID FROM Employees WHERE EmpName = 'Rahul');  
ERROR 1054 (42S22): Unknown column 'EmpName' in 'field list'
```

```
mysql> SELECT Name FROM Employees WHERE DeptID = (SELECT DeptID FROM Employees WHERE Name = 'Rahul');
```

Name
Ravi
Rahul

```
2 rows in set (0.02 sec)
```

```
mysql> SELECT EmpName FROM Employees WHERE Salary = (SELECT MAX(Salary) FROM Employees);
```

```
ERROR 1054 (42S22): Unknown column 'EmpName' in 'field list'
```

```
mysql> SELECT Name FROM Employees WHERE Salary = (SELECT MAX(Salary) FROM Employees);
```

Name
Lohith

```
1 row in set (0.00 sec)
```

DeptName	avg_salary
IT	258500.000000

```
1 row in set (0.01 sec)
```

```
mysql> select min(salary) from employees;
```

min(salary)
33000.00

```
1 row in set (0.00 sec)
```

```
mysql> select max(salary) from employees;
```

max(salary)
440000.00

```
1 row in set (0.00 sec)
```

```
mysql> select EmpName,salary from employees where salary >(select avg(salary)from employees);  
ERROR 1054 (42S22): Unknown column 'EmpName' in 'field list'
```

```
mysql> select count(*) from employees;
```

count(*)
6

```
1 row in set (0.00 sec)
```

```
mysql> select avg(DeptID) from employees;
```

avg(DeptID)
2.0000

```
1 row in set (0.01 sec)
```

```
mysql> select avg(salary) from employees where DeptID = 1;
```

avg(salary)
55000.000000

```
1 row in set (0.02 sec)
```

```
mysql> select avg(salary) from employees where DeptID = 2;
```

avg(salary)
33000.000000

```
1 row in set (0.00 sec)
```

```
mysql> select avg(salary) from employees where DeptID = 3;
```

avg(salary)
258500.000000

```
1 row in set (0.00 sec)
```

```
mysql> select count(*) from employees;
```

count(*)
6

```
1 row in set (0.06 sec)
```

```
mysql> select d.DeptName, e.salary from departments as d inner join employees as e on d.DeptID = e.DeptID;
```

DeptName	salary
CSE	49500.00
CSE	60500.00
ECE	33000.00
IT	77000.00
IT	440000.00

```
5 rows in set (0.05 sec)
```

```
mysql> select e.Name, d.DeptName from employees as e right join departments as d on e.DeptID = d.DeptID;
```

Name	DeptName
Ravi	CSE
Rahul	CSE
Sneha	ECE
Anil	IT
Lohith	IT
NULL	MECH

6 rows in set (0.00 sec)

```
mysql> select * from employees where DeptID is NULL;
```

EmpID	Name	DeptID	Salary	ManagerID	Email
105	Priya	NULL	44000.00	102	priya@kluniv.in

1 row in set (0.00 sec)

```
mysql> |
```

```
mysql> SELECT * FROM Employees;
```

EmpID	Name	DeptID	Salary	ManagerID	Email
101	Ravi	1	49500.00	NULL	ravi@kluniv.in
102	Rahul	1	60500.00	101	rahul@kluniv.in
103	Sneha	2	33000.00	101	sneha@kluniv.in
104	Anil	3	77000.00	102	anil@kluniv.in
105	Priya	NULL	44000.00	102	priya@kluniv.in
106	Lohith	3	440000.00	103	Lohith@kluniv.in

6 rows in set (0.02 sec)

```
mysql> select * from department;  
ERROR 1146 (42S02): Table 'practice2.department' doesn't exist
```

```
mysql> select * from departments;
```

DeptID	DeptName
1	CSE
2	ECE
3	IT
4	MECH

4 rows in set (0.00 sec)

```
mysql> select e.EmpName, d.DeptName from employees as e inner join department as d where e.DeptID = d.DeptID;  
ERROR 1146 (42S02): Table 'practice2.department' doesn't exist
```

```
mysql> select e.EmpName, d.DeptName from employees as e inner join departments as d where e.DeptID = d.DeptID;  
ERROR 1054 (42S22): Unknown column 'e.EmpName' in 'field list'
```

```
mysql> select e.Name, d.DeptName from employees as e inner join departments as d where e.DeptID = d.DeptID;
```

Name	DeptName
Ravi	CSE
Rahul	CSE
Sneha	ECE
Anil	IT
Lohith	IT

5 rows in set (0.02 sec)